

1. Front End Engineering – I (ST-1B)

(HTML, CSS + JavaScript Basics)

1. Introduction to Front-End Development

Front-end engineering focuses on the **user interface (UI)** of a website — what users see and interact with in the browser.

Main technologies:

- **HTML** – page structure
- **CSS** – styling and layout
- **JavaScript** – behavior and interactivity

Browser examples include **Google Chrome** and **Mozilla Firefox**.

2. HTML Basics

HTML (HyperText Markup Language) defines **structure and content** of a webpage.

Basic HTML Page Structure

```
<!DOCTYPE html>
<html>
<head>
<title>My Website</title>
</head>

<body>
<h1>Welcome</h1>
<p>This is my webpage.</p>
</body>

</html>
```

Important HTML Tags

Tag	Purpose
<code><h1></code> - <code><h6></code>	Headings
<code><p></code>	Paragraph
<code><a></code>	Hyperlink
<code></code>	Image
<code><div></code>	Container
<code></code>	Unordered list
<code></code>	Ordered list
<code><table></code>	Table

Example:

```
<ul>
<li>HTML</li>
<li>CSS</li>
<li>JavaScript</li>
</ul>
```

3. CSS Basics

CSS (Cascading Style Sheets) controls **appearance of webpages**.

Example

```
body{
background-color: lightblue;
}
```

```
h1{
color: darkred;
text-align:center;
}
```

Types of CSS

1. Inline CSS

```
<p style="color:red;">Hello</p>
```

2. Internal CSS

Inside `<style>` tag.

3. External CSS

```
<link rel="stylesheet" href="style.css">
```

CSS Box Model

Every element consists of:

- Content
- Padding
- Border
- Margin

Diagram idea students should remember for exams.

4. JavaScript Basics (Important for ST1B)

JavaScript adds **logic and interactivity**.

JavaScript in HTML

```
<script>  
alert("Hello Student");  
</script>
```

Variables

Variables store data.

```
let name = "Student";  
let age = 20;
```

Types:

- `var`
 - `let`
 - `const`
-

Data Types

Type	Example
Number	10
String	"Hello"
Boolean	true
Object	{name:"A"}

Operators

Arithmetic operators

+ - * /

Example

```
let a=10;  
let b=5;  
let c=a+b;
```

Conditional Statements

```
if(age>18){  
  console.log("Adult");  
}  
else{  
  console.log("Minor");  
}
```

Functions

```
function greet(){  
  alert("Welcome!");  
}
```

```
greet();
```

DOM Manipulation

DOM = Document Object Model

Example:

```
document.getElementById("title").innerHTML="New Title";
```

This allows JavaScript to **modify HTML elements dynamically**.